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Geomagnetic Polarity Reversals in a Late Tertiary Lava Sequence from the Akaroa Volcano, New Zealand

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Summary

Palaeomagnetic measurements and potassium–argon age determinations have been performed on two sequences of lava flows from the Banks Peninsula, New Zealand.

Flows from the Stoddart Formation are entirely of reversed polarity and were formed rapidly at $5{\cdot}85 \pm 0{\cdot}04$ My ago. The Akaroa sequence, extruded over a time interval from 9·1 to 8·4 My ago, records seven reversals of the Earth's magnetic field. The geomagnetic polarity time scale constructed from these results agrees well with the time scales postulated from the interpretation of magnetic anomaly patterns over mid-ocean ridges. The age of the polarity reversal terminating the normal period associated with Anomaly 5 is $8{\cdot}67 \pm 0{\cdot}13$ My.

1. Introduction

The Geomagnetic Polarity Time Scale has been well established for the past 3·6 My by combined potassium–argon and palaeomagnetic investigations on volcanic rocks (McDougall & Chamalaun 1966; Cox & Dalrymple 1967, Cox, Doell & Dalrymple 1968). This time scale has been extended to about 80 My ago by Heirtzler *et al.* (1968) by the interpretation of magnetic anomaly patterns observed across mid-ocean ridges using the hypothesis of Vine & Matthews (1963). This extrapolation assumes a constant rate of sea-floor spreading.

The extension of the time scale based on the polarity and age of surface rocks would provide an independent assessment of the history of reversal of the geomagnetic field. Rates of sea-floor spreading could then be measured thus testing Heirtzler's assumption. However, Cox & Dalrymple (1967) and McDougall & Chamalaun (1966) have noted that the precision of potassium–argon dating decreases with age and consequently adequate definition of the ages of the magnetic reversals is extremely difficult beyond about 6 My ago. Attempts to expand the time scale in this way have achieved only limited success (Dalrymple *et al.* 1967; Baksi, York & Watkins 1967; McDougall & Chamalaun 1969; Sasajima 1969).

It is clear that detailed studies on stratigraphic successions of lava flows, rather than on isolated samples, are necessary. Although the replication of measurements yields greater experimental precision, there is no method of assessing the geological reliability of a potassium–argon age determination on an isolated sample. The analysis of samples in a strict stratigraphic order will aid recognition of those samples which have either lost radiogenic argon or which contained excess argon at the time of their formation.

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Successions of lava flows suitable for this type of study occur in the two volcanoes forming the Banks Peninsula, New Zealand (Latitude 43° 40'S, Longitude 172° 45'E). A preliminary survey of the ages of the volcanoes has already been published (Stipp & McDougall 1969). In view of this recent work, together with the detailed geological descriptions of Speight (see Speight (1944) for list of references) and Liggett & Gregg (1965), only a brief outline of the geology and geochronology will be presented here.

Banks Peninsula comprises two volcanoes, Akaroa and Lyttelton (Fig. 1), which are built mainly of mildly alkaline basaltic to andesitic lavas (Speight 1924; Coombs 1963) interbedded with an approximately equal amount of pyroclastics. Stipp and McDougall's data show that Lyttelton was active during the time 11·9 to 9·7 My ago, whilst the lavas forming the main dome of Akaroa were erupted from 9·0 to 8·0 My ago. Volcanic activity did not cease 8 My ago because later lavas were erupted in the vicinity of Mount Herbert and flowed down into the Lyttelton erosion caldera. These lavas comprise the Diamond Harbour Group which was divided into three formations by Liggett & Gregg (1965). The Church Formation was erupted between 8·0 and 7·5 My ago; there is some doubt whether it is continuous with or part of the Akaroa volcanism. The few flows of the Kaioruru Formation were erupted around 6·9 My ago, and these are overlain by the Stoddart Formation, the age of which is 6·0 My.

There is no evidence in the Banks Peninsula of any tectonic events later than the volcanism. Speight (1924) states that no major faulting or folding is to be found. Since considerable care was taken to collect samples from sites where there was no possibility of a local land slip, the palaeomagnetic directions measured are unlikely to have been affected by either local or regional tectonic movements.

2. Sampling sites

Two successions of lava flows were sampled, one in the Akaroa volcano and one in the Stoddart Formation of the Diamond Harbour Group; their locations are shown in Figs 1 and 2.

(i) *Akaroa.* Seventy-three flows were sampled from the Akaroa volcano over a height of 2250 feet (685 m). The lavas exposed in the walls of the erosional caldera were sampled along the Lighthouse Road south of Akaroa village. The lowest 840 feet (256 m) of the section were in a stream bed ascending from the shore of Akaroa Harbour at Green Point. The combined thickness of the ten flows forming this part of the section comprises only about a quarter of the total height. This paucity of rock suitable for sampling is partly due to the poor exposure and is partly a reflection of the composition of the lava pile. Speight (1944) observed that the lower flows around the harbour are interbedded with more than their own thickness of agglomerate and ash.

No lava flows were found in the 300 feet (100 m) of section between the top of the stream and the road, but apart from one small break from 1170 to 1260 feet (357–384 m), the exposure was then almost continuous to 2170 feet (661 m), the road making a notch in the hillside.

Stipp & McDougall (1969) had already dated a sample from near the top of the sequence (8·18 ± 0·24 My, GA 2021), and a sample from sea level at Wainui, across the harbour from Akaroa, gave a minimum age of 9·01 My (GA 2009). Therefore, this section was sampled in the hope that it would represent a time interval of about one million years.

(ii) *The Stoddart Formation.* Fifty-three lavas were sampled in a cliff section on the west side of Purau Valley (Fig. 1). The ten flows at the base were sampled north of the Monument above the head of Purau Bay, the remainder of the section was in a gully 1·5 miles (2·5 km) to the south. In the upper part of the section, above an

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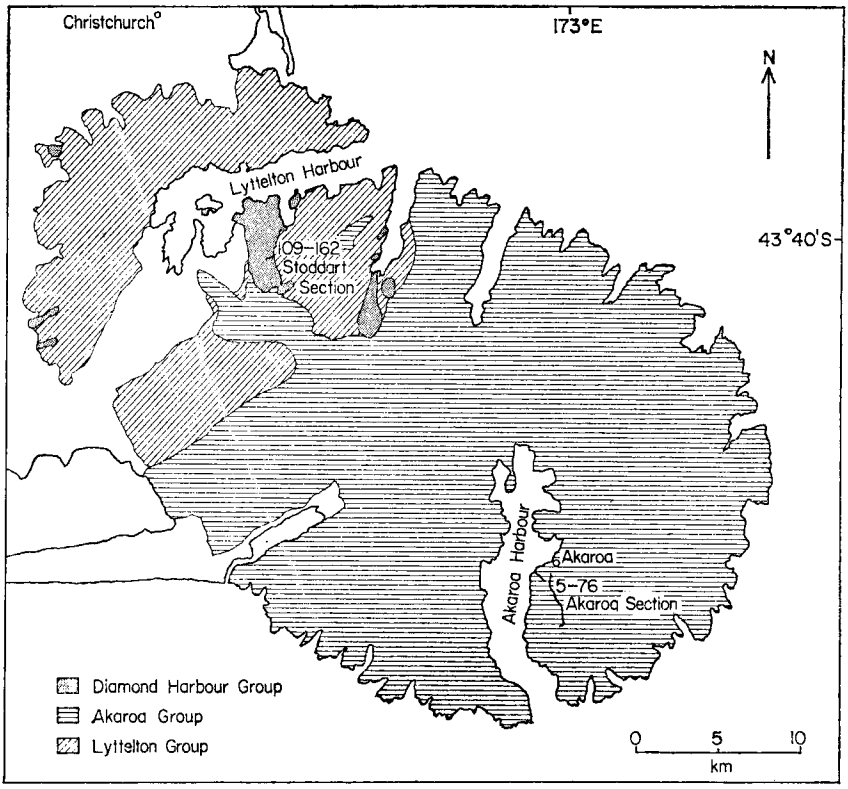


FIG. 1. A geological map of the volcanics of the Banks Peninsula showing the locations of the sampling sites. The numbers adjacent to the sections correspond to the flow numbers in the text.

altitude of about 1400 feet (427 m), the flows are very thin and it was often difficult to distinguish flow boundaries from flow breccias. The vertical extent of the whole section was 900 feet (274 m).

The only control over the age of the base of the section was an age of $8{\cdot}02 \pm 0{\cdot}20$ My (GA 2026) for the underlying Church Formation at Purau Bay. The top flow, BP 152, was dated by Stipp & McDougall (1969) at $5{\cdot}84 \pm 0{\cdot}15$ My (GA 2033). Stipp & McDougall found no age difference in the three samples dated from the Stoddart Formation in the Diamond Harbour area, and there was no indication in the field of much weathering of flow surfaces, so that the lavas of the Stoddart Formation could have been erupted over a short period of time.

3. Geochronology

3.1 *Methods and errors*

All the dates were performed on whole rock samples which showed little or no effect of alteration in thin section. A brief petrographic description of the dated samples is given in Appendix 2. The samples were prepared for argon analysis by crushing and retaining the $-\frac{3}{16}+\frac{1}{8}$ inch fraction. A representative portion of this fraction was then crushed to -100 mesh size for potassium analysis. Ten to fifteen grams of sample were used in each argon analysis.

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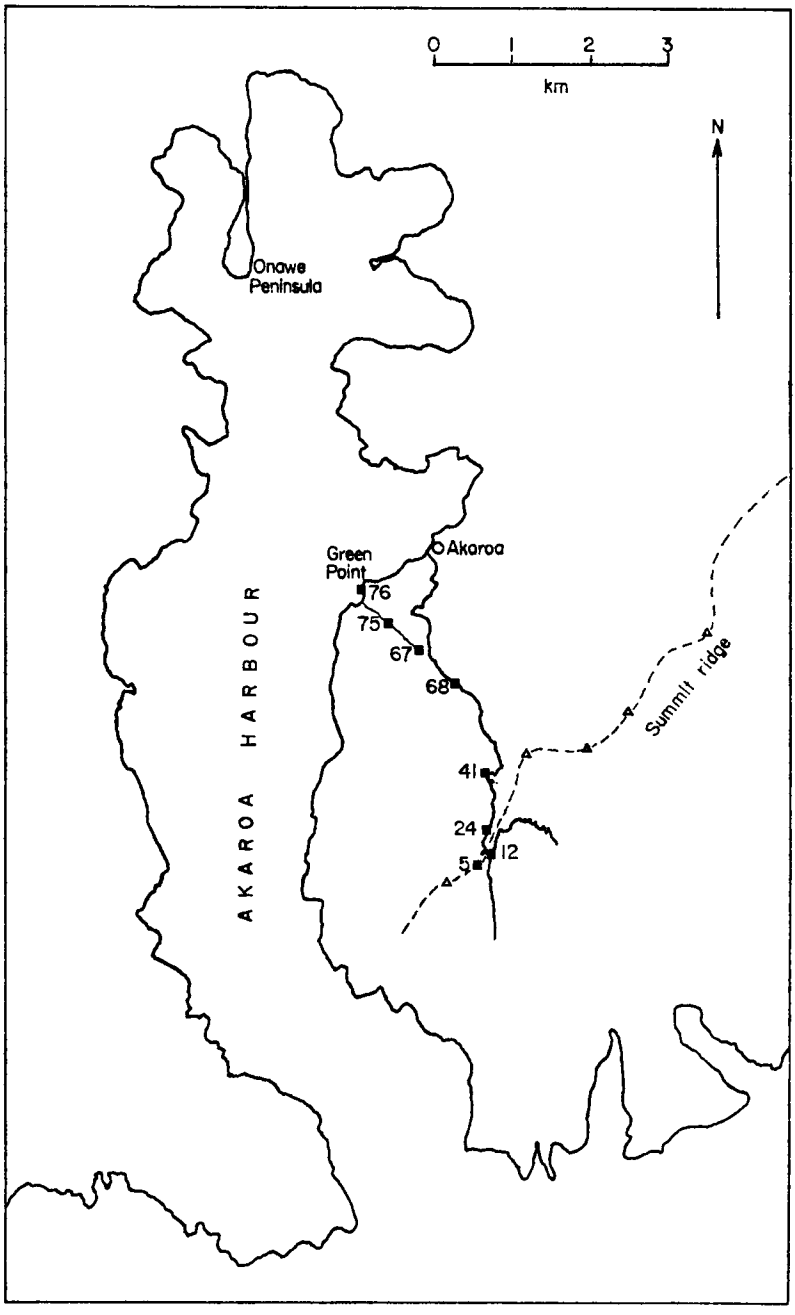


FIG. 2. The location of the Akaroa sequence showing its relation to the present harbour and to the summit ridge of the erosion caldera. The numbers marked on the section correspond to the flow numbers in the text.

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The analytical techniques used for the potassium–argon dating were described in detail by McDougall (1964, 1966) and Cooper (1963). Potassium was measured by flame photometry. The radiogenic argon content was measured by the isotope dilution method using a Reynolds-type mass spectrometer in the static mode. During an argon analysis the isotopic peak heights change with time. The peak heights were fitted to a straight line or to an exponential curve by a least-squares method thus enabling the extrapolation of the values to zero time. Orifice and discrimination corrections were applied to the zero time values of the ratios. Constants used for the calculation of the ages were:

$$\lambda_e = 0{\cdot}585 \times 10^{-10}\ \mathrm{yr}^{-1}.$$

$$\lambda_\beta = 4{\cdot}72 \times 10^{-10}\ \mathrm{yr}^{-1}.$$

$$^{40}\mathrm{K} = 1{\cdot}19 \times 10^{-2}\ \text{atom per cent.}$$

The experimental precision of each age measurement has been estimated by a statistical appraisal of each analysis (McDougall, Polach & Stipp 1969). The final error is a combination of the following three independent errors:

(i) The experimental error in the determination of the ratio of radiogenic ^{40}Ar to tracer ^{38}Ar in each analysis.

(ii) The error associated with the calibration of the volume of ^{38}Ar tracer. This is found by replicate analysis of a standard sample.

(iii) The error in the potassium determination which may be estimated by replicate analysis.

The statistical error associated with each age is reported as one standard deviation in Tables 1 and 2.

An independent evaluation of the precision of an argon determination may be made by the analysis of variance from replicate measurements. The results quoted in Tables 1 and 2 indicate that for a single argon analysis the pooled coefficient of variation is 1·33 per cent. The pooled coefficient of variation for the potassium measurements as determined by replicate analysis is 0·31 per cent. Consequently the combined coefficient of variation for an age determination is 1·37 per cent. It will be noted that the pooled coefficient of variation for an age compares closely with the standard deviation calculated directly for each age determination.

Possible systematic errors in the potassium and argon analyses are difficult to evaluate. Interlaboratory comparisons suggest that the ages are accurate to 2 per cent (S.D.) as shown for example by Lanphere & Dalrymple (1967) using the U.S.G.S. muscovite P–207.

3.2.1 *Potassium–argon age determinations on samples from the Akaroa succession*

The results of potassium–argon age determinations on samples from the Akaroa succession are shown in Table 1. The mean age shown in Table 1 is a weighted mean of the replicate age determinations. The error is the standard error of the mean. The mean age of each flow has been plotted against its height above sea level in Fig. 3.

The advantage of sampling a well-defined succession is illustrated by the age determination of 7·4 My for sample 69–274. This sample has clearly not retained its radiogenic argon thus giving a young apparent age; it has not been plotted in Fig. 3. The only unusual feature of the thin section was the presence of some primary zeolite in the groundmass. The large scatter in the argon determinations of both samples 69–279 and 69–296 is not understood. Since a Jones-type microsplitter was used in preparing each sample for argon extraction, sample inhomogeneity is an unlikely

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Table 1

Potassium–argon ages of samples from the Akaroa sequence

Sample number	Potassium analyses and mean (wt per cent)	Radiogenic argon ($\times 10^{-11}$ mole g^{-1})	Atmospheric argon–40%	Calculated age (My)	Mean age (My)
69–261 BP009	1·163, 1·165 → 1·164	1·669	22·7	8·05 ± 0·14	8·28 ± 0·08
		1·722	30·3	8·30 ± 0·06	
69–262 BP010	1·768, 1·750 → 1·759	2·668	4·8	8·52 ± 0·06	8·49 ± 0·02
		2·654	5·9	8·47 ± 0·06	
69–274 BP012	1·013, 1·016 → 1·015	1·349	21·0	7·47 ± 0·13	
		1·319	12·6	7·30 ± 0·05	
		1·331	19·8	7·37 ± 0·05	
69–276 BP017	1·347, 1·341 → 1·344	1·994	22·8	8·33 ± 0·14	8·42 ± 0·04
		2·019	11·8	8·43 ± 0·06	
69–278 BP021	1·288, 1·284 → 1·286	1·961	10·6	8·56 ± 0·15	8·64 ± 0·03
		1·982	8·2	8·65 ± 0·06	
69–279 BP022	1·289, 1·294 → 1·292	1·841	18·8	8·00 ± 0·14	8·53 ± 0·22
		2·039	22·7	8·86 ± 0·06	
		1·908	17·6	8·29 ± 0·06	
69–280 BP024	1·239, 1·235 → 1·237	1·827	23·7	8·29 ± 0·14	8·25 ± 0·01
		1·818	5·4	8·25 ± 0·06	
69–281 BP026	1·036, 1·037 → 1·037	1·573	21·1	8·52 ± 0·06	
69–282 BP030	1·350, 1·349 → 1·350	2·045	24·9	8·51 ± 0·15	8·33 ± 0·07
		1·996	19·4	8·30 ± 0·06	
69–283 BP032	1·234, 1·224 → 1·229	1·782	23·4	8·14 ± 0·14	8·27 ± 0·05
		1·815	26·9	8·29 ± 0·06	
69–284 BP037	1·112, 1·109 → 1·111	1·707	24·1	8·63 ± 0·07	

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Table 1 (*continued*)

69–285 BP038	1·223 1·221 } 1·222	1·870	23·2	8·59 ± 0·15	
69–287 BP049	1·533 1·545 } 1·539	2·350	19·3	8·57 ± 0·07	
69–288 BP050	1·763 1·759 } 1·761	2·708	18·9	8·63 ± 0·15	
69–294 BP061	2·071 2·084 } 2·078	3·238	14·4	8·75 ± 0·15	
69–296 BP063	0·946 0·953 } 0·950	1·578 1·428 1·504	81·1 58·8 60·4	9·33 ± 0·18 8·44 ± 0·07 8·89 ± 0·12	8·63 ± 0·21
69–298 BP066	1·184 1·184 } 1·184	1·817 1·821	64·1 64·1	8·61 ± 0·08 8·63 ± 0·10	8·62 ± 0·01
69–299 BP068	1·214 1·224 } 1·219	1·878 1·935	63·0 69·5	8·65 ± 0·07 8·91 ± 0·11	8·73 ± 0·12
69–301 BP070	1·502 1·494 } 1·498	2·406 2·452	11·7 19·0	9·01 ± 0·06 9·19 ± 0·07	9·09 ± 0·08
69–303 BP072	1·466 1·474 } 1·470	2·354 2·325	9·4 10·2	8·99 ± 0·16 8·88 ± 0·07	8·90 ± 0·04
69–304 BP074	1·256 1·245 } 1·251	2·045 2·048	37·0 27·4	9·18 ± 0·16 9·19 ± 0·07	9·19 ± 0·01
69–306 BP076	0·868 0·871 } 0·870	1·401	65·0	9·04 ± 0·16	

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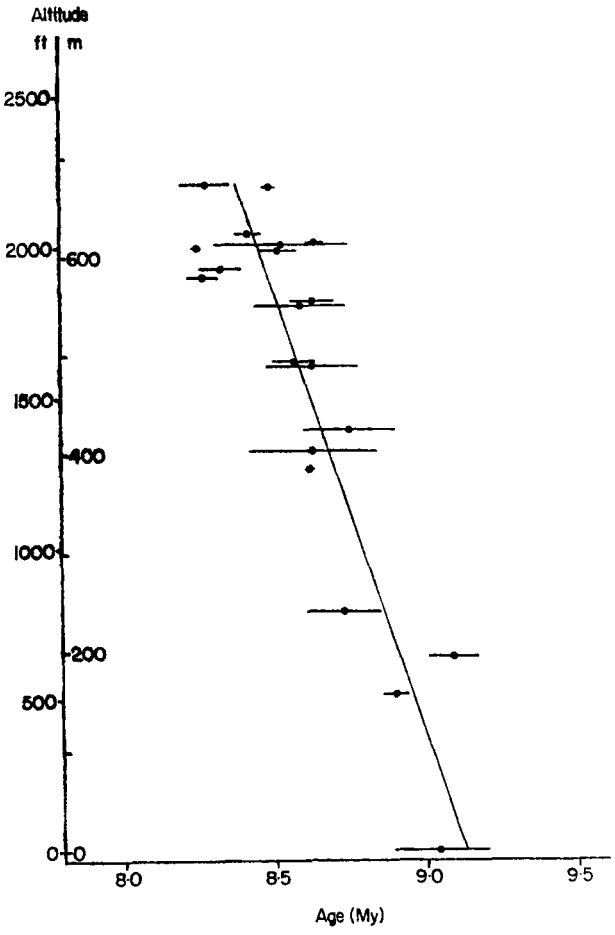


FIG. 3. Potassium–argon age determinations and the altitudes of flows from the Akaroa sequence.

Table 2

Potassium–argon ages of samples from the Stoddart sequence

Sample number	Potassium analyses and mean (wt per cent)	Radiogenic argon ($\times 10^{-11}$ mole g^{-1})	Atmospheric argon-40 per cent	Calculated age (My)
69–323 BP152	1·020, 1·024 | 1·022	1·072	46·5	5·89±0·11
		1·089	23·3	5·99±0·04
69–272 BP129	1·014, 1·015 | 1·015	1·075	74·9	5·95±0·08
		1·037	40·9	5·74±0·04
69–321 BP128	1·028, 1·021 | 1·025	1·058	28·0	5·80±0·04
69–317 BP111	0·836, 0·831 | 0·834	0·868	23·2	5·85±0·04
		0·864	33·4	5·82±0·04
69–316 BP109	0·876, 0·877 | 0·877	0·935	51·7	5·99±0·11
		0·880	44·1	5·64±0·04
		0·910	19·3	5·83±0·04

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explanation. No unusual difficulty was experienced in cleaning the gas samples in the argon extraction lines, and the mass spectrometer analysis gave no indication of impure argon samples. There is no physical reason for rejecting these determinations, hence they have been included in all the data analysis.

It is obvious from Fig. 3 not only that samples from the lower part of the section give older ages than those from the upper part but that there is a general correlation of age and height in the stratigraphic sequence. Let it be assumed that the correlation is linear. The mean age of each flow and its height have been fitted to a straight line using a least-squares method. The resulting line is also shown in Fig. 3. Each flow has been given unit weight irrespective of the experimental error in the age determination. This is because the quoted error includes no estimate of the geological reliability of the age, that is, it gives no indication whether the sample has retained all the radiogenic argon accumulated since its formation.

The equation of the best-fitting straight line is

$$\text{Age} = (9{\cdot}13 \pm 0{\cdot}07) - (3{\cdot}36 \pm 0{\cdot}41)\ 10^{-4} \times \text{height} \qquad (1)$$

where the age is given in My, and the height of the lava flow in feet above sea-level. The errors quoted are standard errors.

The standard deviation of the age of each flow, determined from the deviation of the experimental points from the line, is 0·13 My. The over-all precision for a single age has been estimated to be 1·37 per cent, or about 0·12 My when the age is approximately 9 My. These values cannot be compared with statistical rigour because some of the points forming the line were single age determinations and some were the mean of replicate determinations. Nevertheless, since the deviation is similar to the expected deviation resulting from experimental error, this suggests that the assumption of linearity between age and height is a surprisingly good one. It may be concluded that there are no significant hiatuses in the eruptive history and that the samples whose ages were plotted in Fig. 3 have lost little or no argon since their formation. Having arrived at this conclusion, various methods of weighting the age determinations were attempted. No system of weighting was found which would change any age derived from the line by more than 1 per cent or which would change significantly the deviation of the experimental points from the line.

The best estimate of the age of the lava flow at sea level is $9{\cdot}13 \pm 0{\cdot}13$ My. The best estimate of the age of the highest flow (2210 feet or 674 m) is $8{\cdot}38 \pm 0{\cdot}13$ My. The duration of the volcanism as recorded by the lava succession is inferred to be $0{\cdot}75 \pm 0{\cdot}18$ My. Speight (1944) estimated the original height of the Akaroa volcano to be 5000 feet (1·5 km). The corresponding volume of a cone of 15 km radius is 350 km^3. The rate of eruption was therefore about 5×10^5 m^3 yr^{-1}. This is two orders of magnitude smaller than the recent rates of eruption of Kilauea (10^8 m^3 yr^{-1} since 1923) and Mauna Loa (2×10^7 m^3 yr^{-1} since 1832) (data from Stearns 1966).

3.2.2 *Potassium–argon ages on samples from the Stoddart sequence*

The results of potassium–argon age determinations on samples from the Stoddart sequence are given in Table 2. The age determinations have been plotted against height in Fig. 6 which is shown with the discussion of the palaeomagnetic results in Section 4.2.2. No age difference can be seen between flows from the top and bottom of the sequence. The mean of all the age determinations is $5{\cdot}85 \pm 0{\cdot}04$ My. The error is the standard error of the mean.

4. Palaeomagnetism

4.1 *Methods of analysis*

Two samples were collected from each lava flow and oriented by means of a sun compass and level. In a few cases, samples were oriented using geographic co-ordinates. Samples from the same flow were usually collected 3–5 m apart. In the

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laboratory at least two cores were drilled from each sample and specimens were cut from these cores.

Magnetic measurements were made with a short-period astatic magnetometer (Irving 1956). Systematic errors in measurement are thought not to exceed two degrees.

One specimen from each flow was demagnetized progressively in peak alternating fields of 50, 100, 200 and 300 oersteds and sometimes higher using an alternating field demagnetization apparatus similar to that described by McElhinny (1966). When the direction of magnetization did not change significantly over three successive steps, it was considered that a stable end point has been reached. In general, the unstable components of magnetization were removed after the 100 oersteds demagnetization. In the cases where a stable end point was reached, one more specimen from the sample and two specimens from the other sample from the same flow were demagnetized at the preferred demagnetizing field. The direction of magnetization of each flow is thus the mean of four specimens, two specimens from each of two samples. Specimens exhibiting no stable end point as defined above comprised 5 per cent of the total and these directions were not considered further.

4.2 *Results*

The mean directions of magnetization for both the Akaroa and Stoddart successions are shown in Figs 4 and 5. A statistical analysis of the results is presented in Table 3.

4.2.1 *The Akaroa succession*

The upper part of the Akaroa section records a reversed period and a well-defined normal event. The lower part of the section covers a normal period with a reversed event recorded in two flows near the bottom of the section. The ages of these polarity changes are discussed in Section 4.3.

The dating shows that the succession was formed over a period of 0·75 My. This time interval is two orders of magnitude longer than the 10^4 year period associated with geomagnetic secular variation (Cox 1969a). An attempt has therefore been made to estimate the ancient secular variation in New Zealand by analysing the dispersions of the flow directions measured. Table 3 shows that the angular standard deviation of the flow directions about the overall mean direction is 21·8°. This dispersion is composed of two components, the between-site dispersion which represents ancient secular variation and the within-site dispersion which is represented by the precision of the flow mean directions. Since only two samples were collected from each flow, estimation of the within-site dispersion is subject to considerable uncertainty.

Calculation of the within-site dispersion and the between-site dispersion by the method of Watson & Irving (1957) gives values of 9·8° and 18·4° respectively. The between-site dispersion still contains a small component due to the local magnetic anomalies that existed at the time of extrusion; a correction for this effect would probably be about 1 per cent.

Assuming that the two-level method gives a good estimate of the within-site dispersion, the value of the ancient secular variation is 18·4°. Ninety-five per cent confidence limits of 16·5° and 20·9° can be placed on this result (Cox 1969c, number of flows = 70). In a recent experiment designed to measure the ancient secular variation in New Zealand during the past 0·68 My, Cox (1969b) obtained a value of 11·56° with 95 per cent confidence limits of 9·5° and 14·4°. This suggests that the secular variation of the geomagnetic field in New Zealand may have changed with time.

The greatest angular standard deviation of dispersion for latitude 40°S predicted by models of the change of secular variation with latitude (Cox 1962; Creer 1962;

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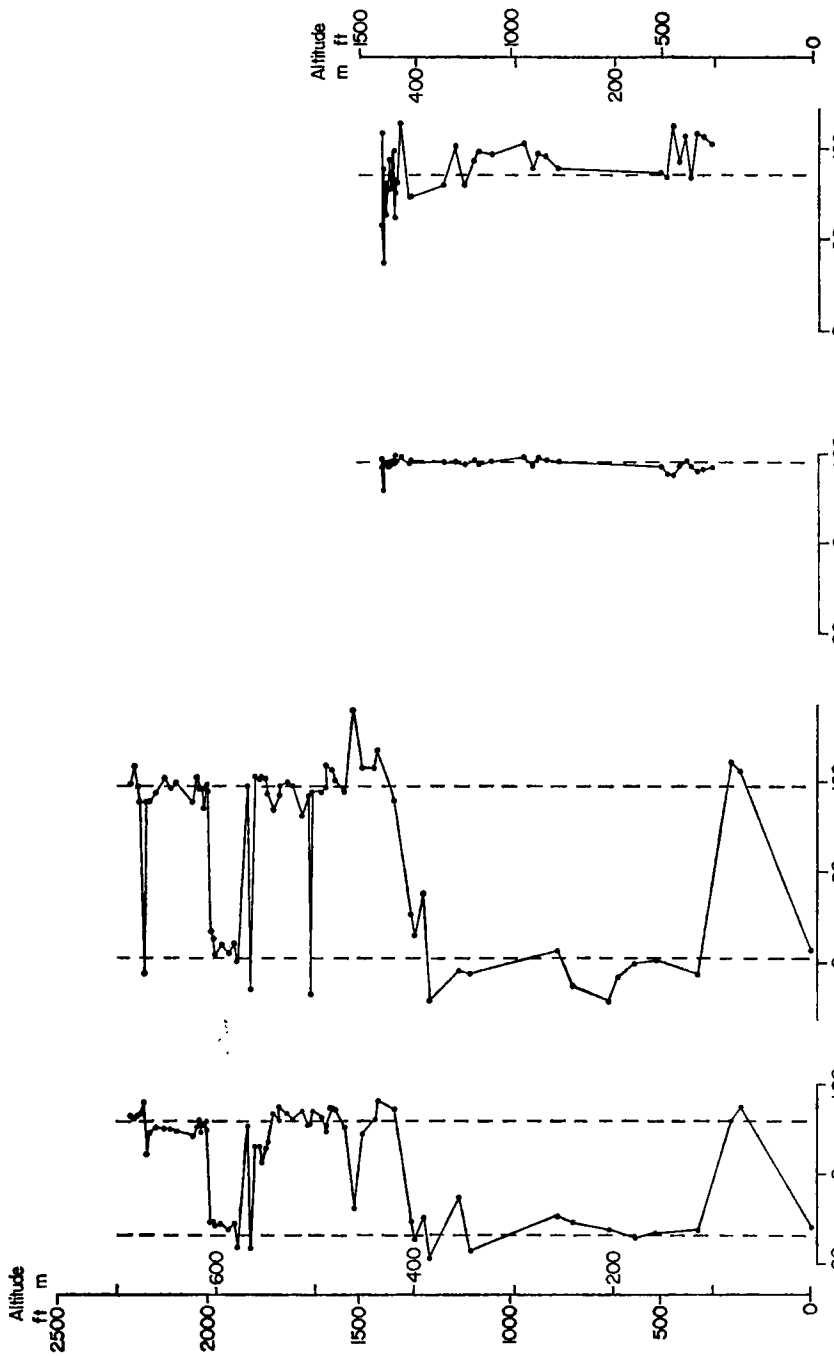


FIG. 4. A stratigraphic plot of the flow mean directions of magnetization for (a) the Akaroa and (b) the Stoddart sequences.

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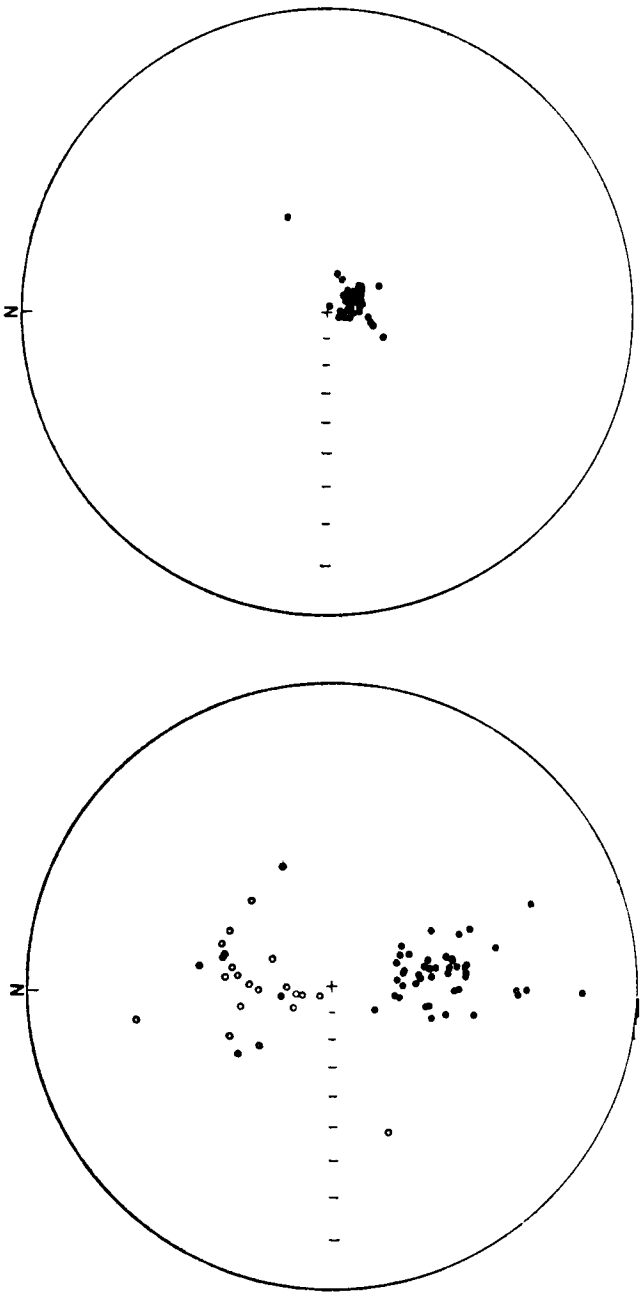


FIG. 5. A stereographic plot of the flow mean directions of magnetization for (a) the Akaroa and (b) the Stoddart sequences. Open circles, upward directions. Closed circles, downward directions.

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Table 3

Statistical analysis of palaeomagnetic results and mean directions

							Mean direction		Corresponding pole	
		N	R	K	α_{95}	δ	Decl.	Incl.	Long.	Lat.
Akaroa	Normal	23	21·0897	11·5163	9·3	23·5	004·2	−60·7	229·2	86·2
	Reversed	47	44·1436	16·1041	5·3	20·1	174·7	52·1	331·1	−78·0
	All	70	65·0118	13·8327	4·7	21·8	177·3	54·9	338·5	−81·3
Stoddart	Reversed	44	43·5225	90·0565	2·3	8·4	156·1	79·6	189·5	−61·1

Notes: N number of flows
R resultant vector sum of N unit vectors
K between flow precision parameter
α_{95} angle of confidence at 95 per cent probability level
δ between flow angular standard deviation factor

Irving & Ward 1964) is the 15·7° of Creer. This was obtained by averaging the directions of the present geomagnetic field along latitude 40°S. The results discussed in this paper show that in the past, the amplitude of secular variation in New Zealand was greater than that predicted by present models.

The pole position derived from the mean of all flow directions is longitude 338·5°E, latitude 81·3°S. This pole is on the opposite side of the geographic south pole from New Zealand. Unless the geomagnetic south pole was situated in this position 9 My ago, its apparent position could only be produced by a drift of New Zealand towards the South Pole during the last 9 My. Since New Zealand is thought to have drifted away from Antarctica, movement of the geomagnetic pole over the last few million years seems a more likely explanation. The lack of palaeomagnetic data for New Zealand prevents comparison of the New Zealand polar wander path with the well-determined Australian (Wellman, McElhinny & McDougall 1969) or Antarctic (McElhinny & Wellman 1969) curves.

4.2.2 *The Stoddart succession*

This sequence proved to be entirely of reversed polarity. Only one sample was measured from each of 28 flows but two samples were measured from the other 16 flows. The tight grouping and small flow-by-flow change of direction of magnetization suggest that the sequence was extruded in a time interval too rapid to average out secular variation. The potassium–argon results (Table 2 and Fig. 6) are consistent with this interpretation since no age difference can be detected between flows at the top and bottom of the sequence.

The mean age of the five dated flows is $5{\cdot}85 \pm 0{\cdot}04$ My. This appears to correspond well with the reversed interval between 5·85 and 5·76 My in Vine's (1968) time scale, and between 5·96 and 5·88 My in the time scale of Heirtzler *et al.* (1968). If these estimates of the length of the reversed polarity interval are correct, the Stoddart Formation was erupted in a time less than 0·09 My.

4.3 *The Geomagnetic Polarity Time Scale*

The altitudes of the flows between which polarity changes occur in the Akaroa sequence have been listed in Table 4. The corresponding ages calculated from equation (1) are also shown together with an estimate of the age of the polarity

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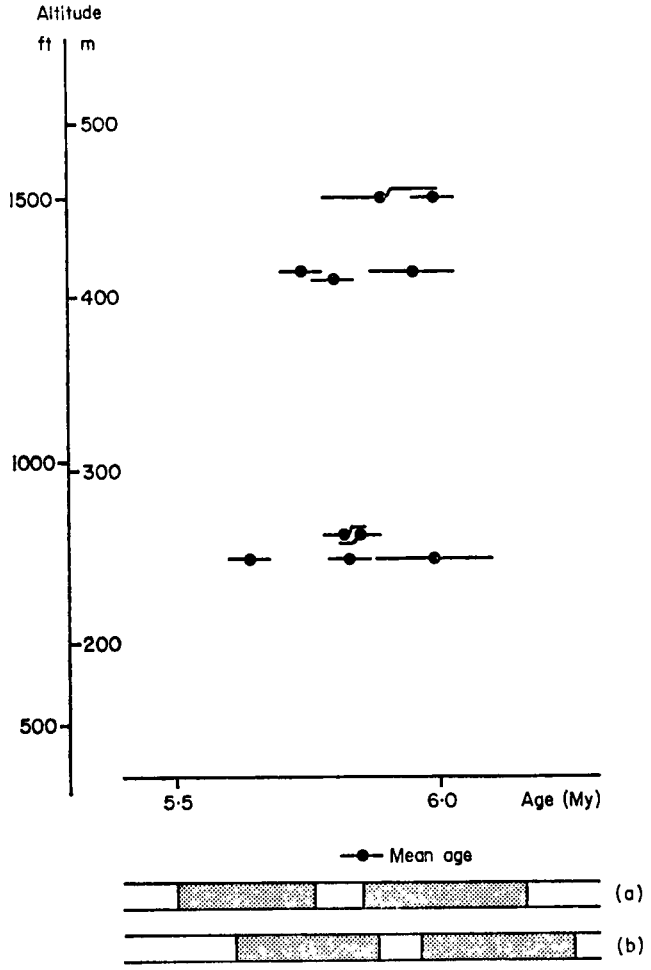


FIG. 6. Potassium–argon age determinations and the altitudes of flows from the Stoddart sequence. The geomagnetic polarity time scales of (a) Heirtzler *et al.* (1968) and (b) Vine 1968 for the same time interval are also shown. Normal polarity intervals are shaded. The Stoddart sequence is entirely of reversed polarity.

Table 4

The altitudes and ages of polarity changes in the Akaroa section

Sense of polarity change Lower Flow—Upper Flow	Limits of altitude (ft) Lower Flow	Limits of altitude (ft) Upper Flow	Corresponding limits of age (My) Lower Flow	Corresponding limits of age (My) Upper Flow	Mean age of polarity change
N — *R*	0	235	9·13	9·05	9·09
R — *N*	265	375	9·04	9·00	9·02
N — *R*	1325	1380	8·68	8·66	8·67
R — *N*	1840	1860	8·51	8·50	8·51
N — *R*	1860	1865	8·50	8·50	8·50
R — *N*	1865	1900	8·50	8·49	8·50
N — *R*	1990	2000	8·46	8·45	8·46

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change. The error in the age quoted for a single polarity change is $\pm 0{\cdot}13$ My; it is of the same order as the error associated with a single age determination from Fig. 3. This error is larger than the uncertainty in the estimation of the altitude in the lava sequence at which the polarity change occurs.

The ages of the polarity changes are shown in Fig. 7 with the time scales of Vine (1968) and Heirtzler *et al.* (1968) derived from the interpretation of the oceanic magnetic anomaly pattern.

Both time scales are compatible with the results from the Akaroa section; the correlation of the polarity intervals is shown in the figure. It is important to realize

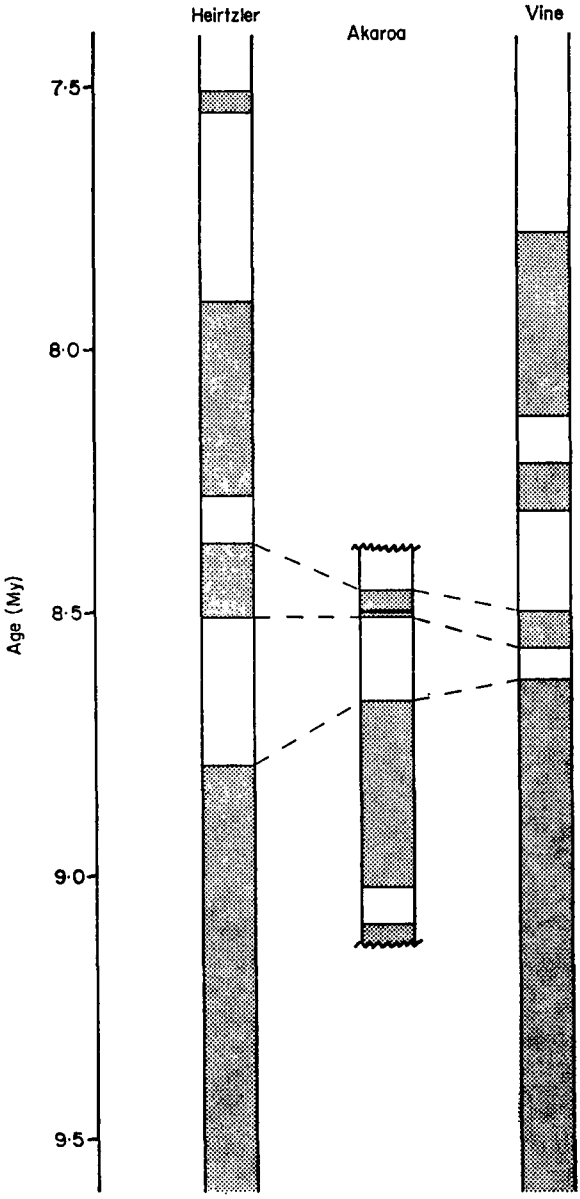


FIG. 7. A comparison of the time scales of Vine (1968) and Heirtzler *et al.* (1968) with the time scale derived from the Akaroa sequence. Normal polarity intervals are shaded.

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that this interpretation is not unique, but the good correlation argues against it being fortuitous. The alternative interpretation that the short normal interval at 8·5 My in Akaroa section corresponds to Heirtzler's 7·5 My normal period implies either a systematic error of 1 My in the Akaroa dating or an error of 10 per cent in the sea-floor spreading rates.

The short reversed event at 9·1 My was recorded in two flows in a poorly exposed part of the section. Any estimate of its duration is therefore subject to a large uncertainty. This event is not recorded in either Heirtzler's or Vine's interpretation, the time interval involved presumably being too small to be observed in the oceanic anomaly pattern. Similarly, the short reversed event at 8·5 My recorded in only one flow in the Akaroa sequence, is also unlikely to be of sufficient time duration to influence the oceanic magnetic anomalies.

It is possible to estimate the age of the polarity reversal terminating the normal period associated with Heirtzler's Anomaly 5. This anomaly is one of the most conspicuous features of the oceanic magnetic patterns and can be recognized in almost all profiles. The various estimates of this polarity reversal may be compared with the Akaroa data:

Vine (1968)	8·63 My
Heirtzler *et al.* (1968)	8·79 My
Akaroa	8·67 My

The main conclusion of this work is that, provided the interpretation shown in Fig. 7 is correct, the published time-scales of geomagnetic reversals and consequently of sea-floor spreading for the past 9·2 My are correct to within about 1 per cent.

If future work invalidates the correlation shown in Fig. 7, the geomagnetic polarity time-scale now established for the period 9·2–8·4 My ago will contribute to the determination of sea-floor spreading rates in the past nine million years.

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Appendix 1

Intensity studies

In order to analyse the intensity of magnetization of samples from Akaroa, the distribution of intensities after demagnetization at a certain field has been assumed to be logarithmic normal (Irving, Molyneux & Runcorn 1966). Normal and reversed samples have been considered separately (Table 5 and Fig. 8).

The curves of mean intensity against peak demagnetizing field (Fig. 8) are similar to those of Irving & Roy (1968), but at high fields the mean intensity of the normal samples is significantly lower than the mean intensity of the reversed samples. This may mean (a) that the time interval of sampling of the geomagnetic field was insufficiently long to make the mean normal and reversed field intensities converge, or (b) that the lava flows which cooled in a reversed field differ petrologically from those which cooled in a normal field, or (c) that the intensity of the Earth's field depends upon its polarity, the intensity being greater at reversed polarity.

It is worth noting that the ratio of the mean intensities for normal and reversed samples stays constant over the higher demagnetizing fields. Since the coercive spectrum is similar for the two cases, this suggests that the location of the remanent magnetization is independent of polarity.

Appendix 2

Locations and petrographic notes on the samples used for potassium–argon dating

Grid references were read from the New Zealand 1:63360 Topographical maps, sheets S84 and S94. Heights above sea-level were measured with an altimeter and

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are thought to be correct to within 5 feet. In the absence of chemical analyses, no attempt has been made to distinguish andesitic basalts from basaltic andesites. Where no comment is made on the petrography of the sample, it may be assumed that the thin section showed a fresh and holocrystalline rock.

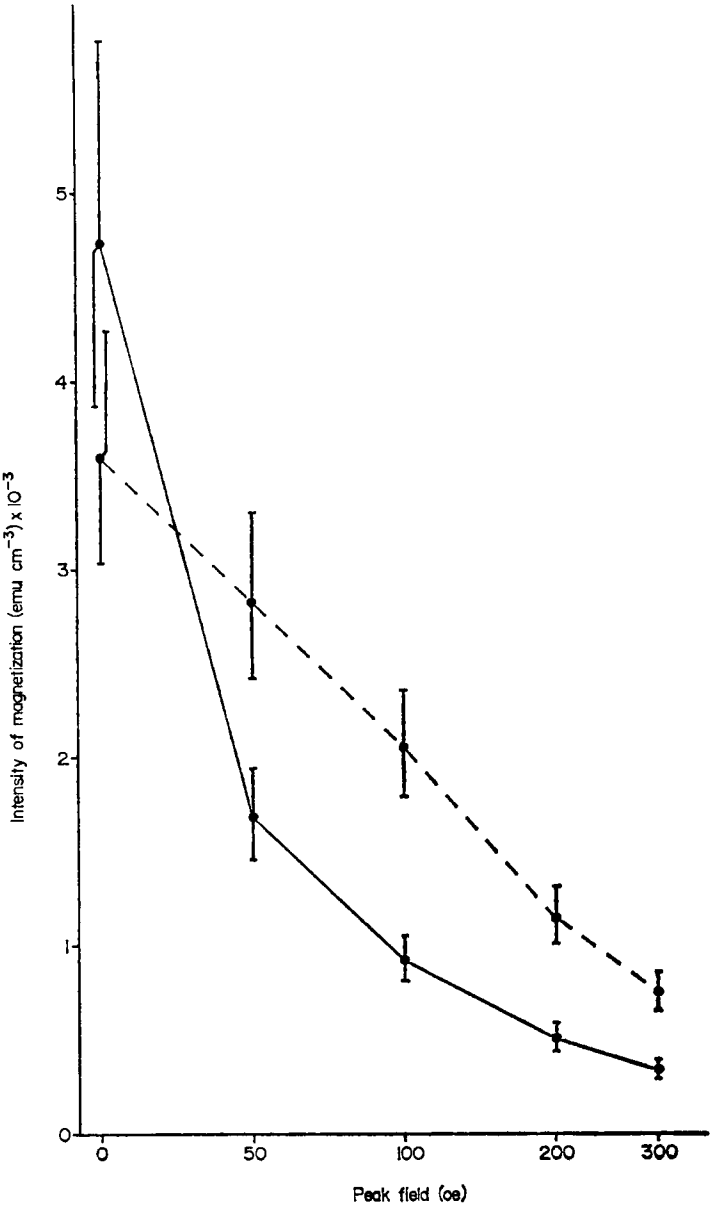


FIG. 8. The effect of a.c. demagnetization of the mean intensities of normal and reversed samples from the Akaroa sequence. Standard errors are shown. The broken line denotes reversed samples.

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Table 5

Summary of the remanent intensities of samples from the Akaroa sequence

Treatment	N	Polarity	Logarithmic means Mean	S.D.	S.E.	Mean Intensity $\times 10^{-6}$ emu cm^{-3}	Standard Error
NRM	23	*N*	3·676	0·423	0·088	4742	+1066 −869
NRM	47	*R*	3·556	0·509	0·074	3597	+669 −563
AC050	23	*N*	3·226	0·295	0·062	1683	+258 −224
AC050	47	*R*	3·452	0·467	0·068	2831	+480 −410
AC100	23	*N*	2·966	0·273	0·057	925	+130 −113
AC100	47	*R*	3·313	0·413	0·060	2056	+304 −265
AC200	23	*N*	2·701	0·308	0·064	502	+80 −69
AC200	47	*R*	3·061	0·397	0·058	1151	+164 −144
AC300	23	*N*	2·532	0·293	0·061	340	+51 −44
AC300	46	*R*	2·873	0·417	0·061	746	+113 −98

Notes: N number of samples

Akaroa section

BP 9 69–261 S94/292168 2210 feet (674 m).

BP 10 69–262 S94/292168 2200 feet (671 m).
Fluidal texture, fine grained.

BP 12 69–274 S94/293169 2170 feet (661 m).
Olivine phenocrysts iddingsitized. Groundmass imperfectly crystallized in places, but little evidence of secondary minerals. Groundmass is 10% zeolite.

BP 17 69–276 S94/292170 2045 feet (623 m).
Very slight amount of glass.

BP 21 69–278 S94/292171 2015 feet (614 m).
Very slight amount of glass.

BP 22 69–279 S94/292171 2010 feet (613 m).
Small amount of fresh isotropic glass.

BP 24 69–280 S94/292171 2000 feet (610 m).
Poorly-defined potassium feldspar in groundmass. The small amount of glass is quite fresh.

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BP 26 69–281 S94/292172 1990 feet (607 m).
Olivine completely iddingsitized, but the rock is otherwise fresh. Groundmass poorly crystallized in places.

BP 30 69–282 S94/293175 1930 feet (588 m).
Groundmass better defined than BP 26. Occasional traces of chlorite.

BP 32 69–283 S94/293175 1900 feet (579 m).
Olivine slightly altered.

BP 37 69–284 S94/294180 1820 feet (555 m).
Olivine almost completely iddingsitized. Some chlorite in the groundmass.

BP 38 69–285 S94/294180 1805 feet (550 m).
Some chlorite in the groundmass.

BP 49 69–287 S94/294184 1620 feet (494 m).
Although most of the rock is quite fresh, there are traces of secondary minerals.

BP 50 69–288 S94/294186 1605 feet (489 m).
Fluidal texture. Olivine occasionally altered. The traces of brown glass seem quite fresh.

BP 61 69–294 S94/292192 1395 feet (425 m).
Some chlorite, sericite and devitrified glass among the fine-grained plagioclase and potassium feldspar of the groundmass.

BP 63 69–296 S94/291193 1325 feet (404 m).
Phenocrysts of slightly altered olivine.

BP 66–1 69–298 S94/291193 1265 feet (386 m).
Occasional patches of devitrified glass and chlorite.

BP 68 69–299 S94/283201 790 feet (241 m).
Olivine slightly altered. Some calcite present.

BP 70 69–301 S94/282202 640 feet (195 m).
Traces of calcite.

BP 72 69–303 S94/281202 515 feet (157 m).
Some calcite and chlorite.

BP 74 69–304 S94/278204 265 feet (81 m).
Tendency towards fluidal texture. Patchy areas of chlorite and calcite.

BP 76 69–306 S94/274212 0 feet
Olivine only slightly altered, but chloritic areas in poorly crystallized groundmass.

Stoddart section

BP 109 69–316 S84/094406 815 feet (248 m).

BP 111 69–317 S84/094406 860 feet (262 m).
Slight alteration of olivine.

BP 128–1 69–321 S84/094403 1345 feet (410 m).
Olivine phenocrysts altered along rims and cracks. Slight traces of alteration in groundmass.

BP 129 69–272 S84/094403 1360 feet (415 m).
Olivine more altered than BP 128–1.

BP 152 69–323 S84/088402 1500 feet (457 m).
Same outcrop as Stipp & McDougall (1969) Sample 37 (GA 2033).

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